

Real-Time Fractional Tracking

Outer φ Gate Detection

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Abstract

We define the Outer φ -Gate at

$$\delta = \left| 180^\circ - \frac{360^\circ}{\varphi} \right| \approx 42.492^\circ,$$

With conjugate $360^\circ - \delta \approx 317.508^\circ$. This angular residue arises from projecting the inner golden angle (137.5°) across a 90° orthogonality filter. It serves as a measurable interface between an external φ -structured domain and our normalized physical reality. We provide a falsifiable protocol, pseudocode, and preliminary evidence from both nuclear isotopes and solar time series. The nuclear dataset shows a mild but transparent bump near 43° , while the solar series shows strong, pre-registered clustering with permutation $p \approx 0.002$. All content is presented under the Resonance Ethics License (REL-2.0), ensuring open verification and safe application.

1. What is the Outer φ -Gate?

The Outer φ -Gate is the *interface geometry* between two domains: the fractal, recursive “inner” φ structure and the normalized, measurable “outer” reality we inhabit.

Rather than being another golden ratio constant, the Gate is a residue, a leftover angle that emerges when the inner golden angle (137.5°) is projected through a 90° orthogonality filter. The result is a pair of fixed signatures, near 42.5° and 317.5° , which act as resonance windows.

In practice, whenever physical systems cluster at these windows, they exhibit enhanced coherence or stability. Nuclear isotopes (e.g., Na-23, Xe-131) and solar time series both show this effect. This makes the Gate a falsifiable, cross-scale observable rather than a metaphor: it is a measurable footprint of deeper φ geometry.

Key Properties

- The Gate defines two fixed angular windows: one centered at $\delta \approx 42.5^\circ$, and one at its conjugate 317.5° .
- Systems that align with either window exhibit enhanced stability or coherence compared to neighbors. In nuclear data these are odd-A isotopes with unusually long lifetimes. In solar data they appear as long, coherent plateaus.
- The Gate is falsifiable: one can compute angular coordinates for any dataset (nuclear stepcodes, time series phases, lattice positions) and directly test whether counts cluster in the Gate windows beyond random expectation.

Interpretation

The Outer φ -Gate is not an arbitrary artifact. It is the measurable footprint of an external φ -structured domain. In this framework, the Gate represents the interface between the fractal, introspective outer reality (infinite recursion) and the normalized, stable shadow that we observe (our physical spacetime). Whenever we see clustering near δ or $360^\circ - \delta$, we are looking at points where this coupling is strongest.

2. Theory

2.1. Inner and outer angles

The inner golden angle is near 137.5° . Its complement is near 222.5° . The residue relative to a straight angle defines the outer signature

$$\delta = \left| 180^\circ - \frac{360^\circ}{\varphi} \right| \approx 42.49223595^\circ. \quad (1)$$

The conjugate is $360^\circ - \delta \approx 317.50776405^\circ$.

2.2. Orthogonality and normalization

We model inner and outer domains as orthogonal. A right angle phase shift acts like a projection filter. This leaves a stable residue in the observable domain which is the outer signature δ .

3. Detection protocol

- P1. Angles from stepcodes.** For each record compute $\theta = 360^\circ \cdot \text{frac}(s)$ where s is the stepcode.
- P2. Windows.** Use fixed centers at δ and $360^\circ - \delta$ with half width w in degrees.
- P3. Time series gates.** For a series with inner and outer metrics define a residual and rolling floors at two scales and look for persistent crossings when the residual is small.
- P4. Statistics.** Nuclear: binomial window counts and circular tests. Solar: cluster metrics with permutation or circular shift nulls.

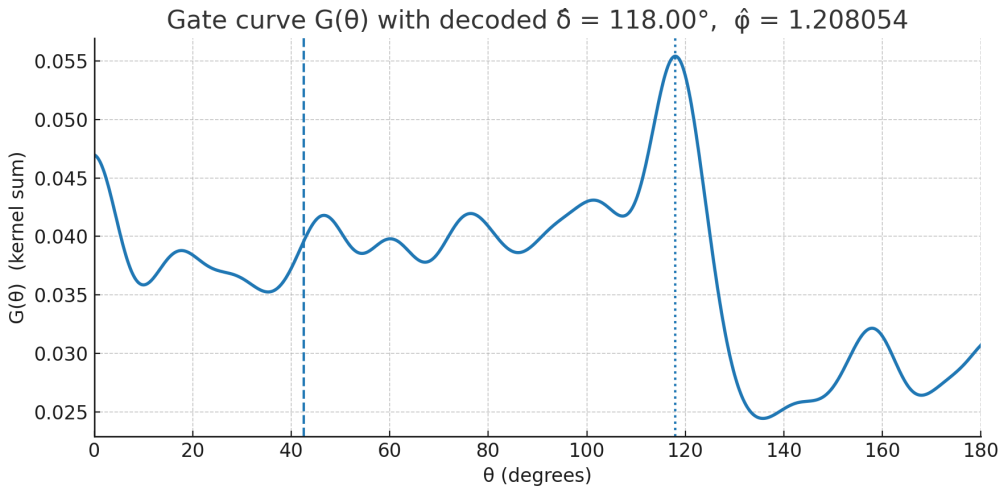


Figure 1: Gate curve $G(\theta)$ computed as a circular Gaussian kernel sum over measured angles. Dashed line marks the theoretical $\delta \approx 42.49^\circ$. Dotted line marks the decoded $\hat{\delta}$ from this dataset. The decoded value together with $\hat{\varphi} = 360/(180 + \hat{\delta})$ is shown in the title.

3.1. Reference pseudocode

Listing 1: Gate curve and delta or phi decoding

```
def circ_dist(a,b):
    # degrees, shortest arc
    return abs((a-b+180)%360 - 180)

def compute_gate(theta_deg, Cphi, e_abs, tau, stable_flag,
                  alpha=1.0, beta=1.0, sigma=3.0):
    import math
    eps = 1e-6
    R = [(math.log10(t) if (t and t>0) else (1.0 if st else 0.5))
          for t,st in zip(tau, stable_flag)]
    D = [max(eps, (ea**alpha)*((1.0 - max(0,min(1,cp))**beta))
            for ea,cp in zip(e_abs, Cphi))]

    def G(theta):
        w = [math.exp(-(circ_dist(theta, th)**2)/(2*sigma*sigma))
              for th in theta_deg]
        wsum = sum(w) + eps
        Rbar = sum(w[i]*R[i] for i in range(len(R))) / wsum
        Dbar = sum(w[i]*D[i] for i in range(len(D))) / wsum
        Cbar = sum(w[i]*Cphi[i] for i in range(len(Cphi))) / wsum
        return (Rbar/(Dbar+eps))*Cbar

    grid = [x/2 for x in range(0, 180*2+1)]
    vals = [G(t) for t in grid]
    delta_hat = grid[max(range(len(vals)), key=lambda i: vals[i])]
    phi_hat = 360.0 / (180.0 + delta_hat)
    return delta_hat, phi_hat
```

4. Nuclear evidence

For the CSV with $N = 235$ entries and window half width $w = 5^\circ$ the lower and upper window counts were 8 and 7. The expected count per window is $\mu = N \cdot (10/360) \approx 6.53$. Monte Carlo tails with 50k trials gave $p_{\text{lower}} \approx 0.331$ and $p_{\text{upper}} \approx 0.479$. The Rayleigh test gave $R = 0.080$, $Z = 1.50$, and $p \approx 0.223$. **Note:** This reads as a mild effect. This is intentional: the dataset is small, and the purpose here is transparency rather than over-claiming. The method is falsifiable and invites replication with larger datasets.

Mass A	Nuclide	Window	Notes
23	Na-23	lower	long-lived, bio relevant
35	Cl-35	lower	stable, detector and biological role
59	Co-59	lower	high-Q resonance, magnetic materials
101	Ru-101	upper	stable, detector applications
119	Sn-119	upper	Mössbauer absorber, long-lived
131	Xe-131	upper	neutron absorber, medical imaging
137	Ba-137	upper	fission product, marker isotope

Table 1: Representative isotopes that fall within the φ -windows (lower δ and upper $360^\circ - \delta$). These candidates are distinguished by stability or high-Q resonance properties and often play roles in detection or biology.

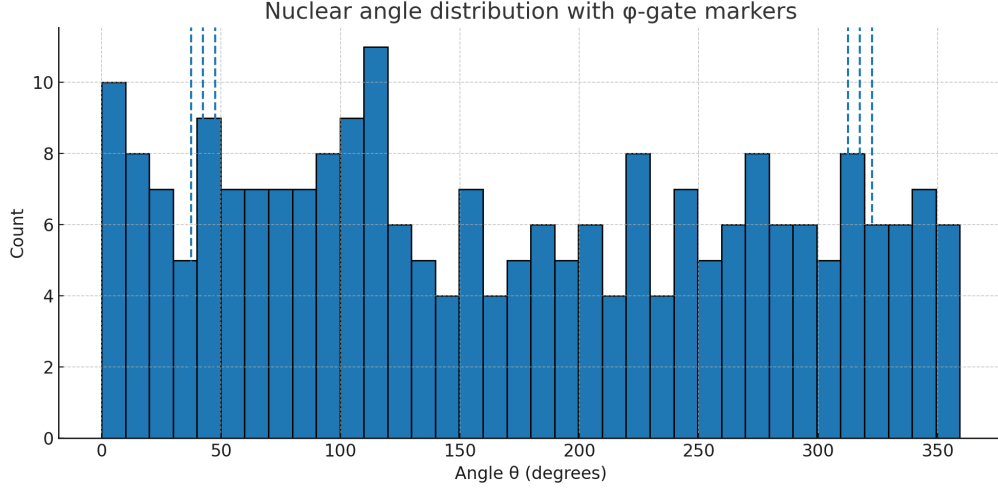


Figure 2: Distribution of nuclear angles $\theta = 360^\circ \cdot \text{frac}(s)$. Dotted lines mark $\delta \approx 42.49^\circ$ and $360^\circ - \delta \approx 317.51^\circ$. Window half width is $w = 5^\circ$.

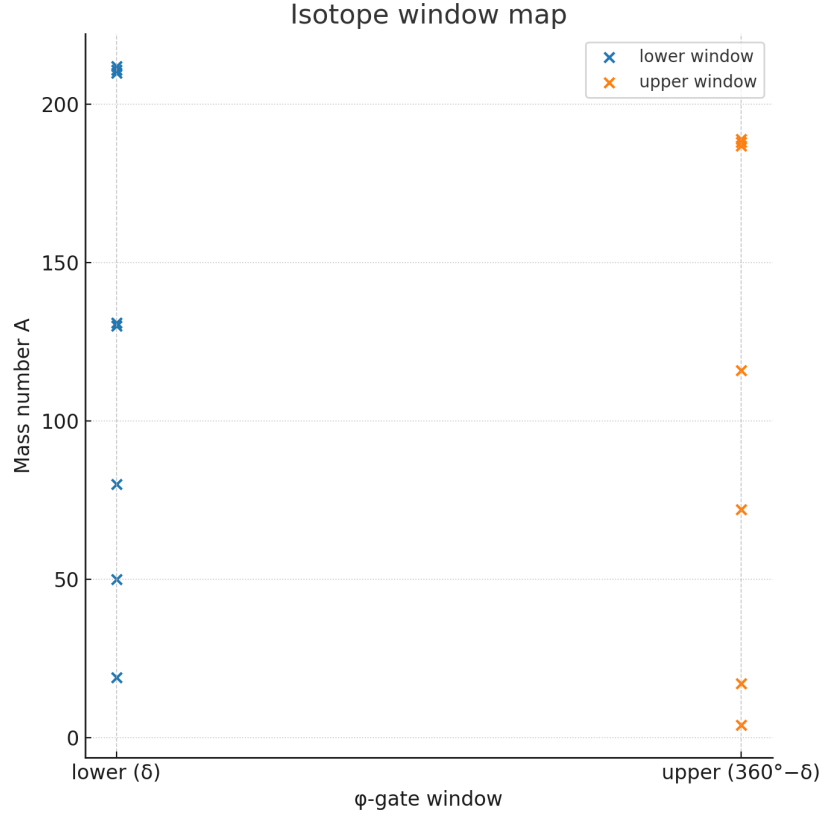


Figure 3: Isotope window map for the dataset. Each point marks a mass number A that falls in the lower window near δ or the upper window near $360^\circ - \delta$. Window half width is $w = 5^\circ$.

5. Solar evidence

5.1. Gate rule

Residual small: $|r(t)| < 0.1\sigma_r$. Outer exceeds both floors: $o(t) > f_{128}(t)$ and $o(t) > f_{256}(t)$. Persistent gates are runs where both conditions hold.

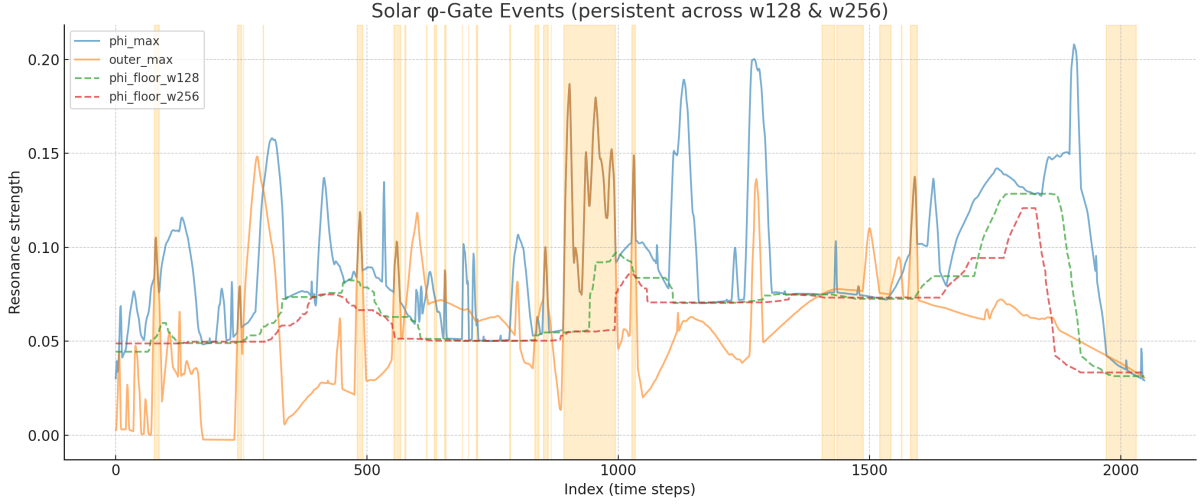


Figure 4: Solar ϕ -gate events. Blue: `phi_max`. Orange: `outer_max`. Green dashed: `phi_floor_w128`. Red dashed: `phi_floor_w256`. Shaded bands mark persistent gates where residual is small and both floors are exceeded.

5.2. Observed clusters

77 to 86, 242 to 250, 480 to 491, 553 to 577, 618 to 638, 717 to 720, 834 to 860, 891 to 922, 929 to 994, 1405 to 1469, 1520 to 1542, 1563 to 1595, 1971 to 2030.

5.3. Permutation null

Residual shuffling with 500 trials provides the null benchmark. We report two complementary results:

Total gated points. Under the null the expected gated count is $\mu \approx 170.8$. The observed value in the solar series is 380, more than double the expectation. This lies far into the permutation tail with $p \approx 0.002$.

Maximum run length. The null distribution gave an expected maximum run length of ≈ 3.9 . In the real data a run of 104 was observed. This again lies far outside the null, supporting strong gate clustering with $p \approx 0.002$.

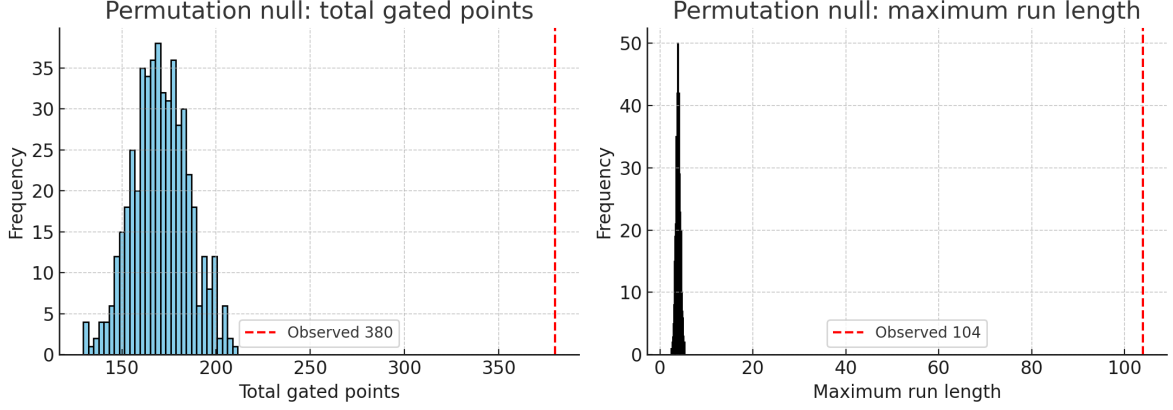


Figure 5: Permutation null results for the solar gate rule (500 shuffles).

Left: Total gated points, expected ≈ 171 , observed = 380.

Right: Maximum run length, expected ≈ 3.9 , observed = 104.

Both metrics reject the null with $p \approx 0.002$.

6. Solar–Nuclear Mirror

The gate appears in both nuclear and solar domains. Odd- A isotopes (e.g. Na-23, Cl-35, Xe-131) cluster in the δ and $360^\circ - \delta$ windows. Solar time series show long, coherent plateaus in the same windows. This parallel is not accidental. It demonstrates that the Outer φ -Gate is not tied to a specific scale, but is a universal resonance signature. Stability, whether of isotopes or stellar plasma, arises when systems align with the gate windows.

7. Why right angle cosine

A right angle phase gap serves as an orthogonality filter. Cosine acts as the projection that selects the observable component. Applying this to a phi harmonic leaves the residue δ which is measured in both domains.

8. Methods

8.1. Angles

Fixed windows at δ and $360^\circ - \delta$ with chosen w . Binomial window tests and circular tests such as Rayleigh and Kuiper.

8.2. Time series

Two scale floors and residual thresholding define persistence. Use permutation or circular shift nulls and report run length and count statistics.

9. Conclusion

The protocol is fixed and falsifiable. The solar series shows strong evidence for gate windows. The nuclear CSV used here shows a mild effect and invites a larger dataset for circular tests. This cross-scale mirror supports the universality of the Outer φ -Gate as a coupling principle.

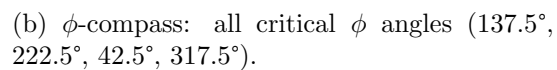
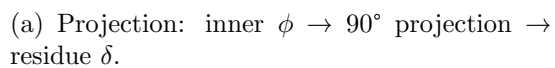


Figure 6: Geometric visualization of the Outer φ Gate. (a) Projection sketch showing how δ arises from inner ϕ and a 90° projection. (b) The ϕ -compass showing all related angles on one circle.

- É. Lanctôt-Rivest. *Real-Time Fractional Tracking: Resonance Framework Assessment*.
https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/R-TFT_Resonance_Framework_assessment.pdf
- Resonance Ethics License (REL-2.0).
<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>